English Communication & Engineering Career Skills

Lecture 5: The Internship Project

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Aims

Look at the Internship Project in detail and how it relates to:

- the Portfolio
- the Group Interview
- the Written Assignment
- the Presentation

You've read the Engineering Internship Post, right? You listened carefully to Lecture 3, right?

But what did you learn about the project connected to the post?

The internship project is a “mini” version of projects run by Praxis IRL

Relates to GC: Restore/improve urban infrastructure

We work with a pool of engineering talent from top universities, research institutions and private companies, alongside a wide range of local and national governments, charities and development funds, to:

You identify **one** issue related to **one locality**

- identify pressing issues affecting specific localities
- research and evaluate solutions, taking into account local conditions
- define and cost the optimum solution
- secure public and private funding
- manage and implement the solution in collaboration with local bodies

You research **two** solutions, evaluate and choose the best one for the locality

The only relevant part here is “define the optimum solution”

More detail on the project

What's this internship about?
Praxis' interdisciplinary teams typically work on projects connected to the *NAE Grand Challenges for Engineering*, and our unique **Young Engineer Summer Research Internship** supports successful candidates to investigate **one** of the challenges in the context of a **specific locality**. For this year's programme, we ask you to focus on the challenge to **Restore and Improve Urban Infrastructure**. That means choosing a **location** (a city or town) with a specific **issue** related to urban infrastructure that you believe can be solved by engineers.

If you are accepted as one of our summer interns, you will work closely with a team to sharpen your research and critical thinking skills, investigate approaches to solving the issue you have identified, and produce a report which recommends the optimum solution. Your idea will then have the chance to become reality!

This should be a place "where factors such as poor infrastructure or a lack of local investment can make it challenging to implement positive change"

Possibly in a developing country?

Otherwise, the issue is not urban!

Solutions need to have a technical engineering aspect – not solved just by laws/regulations

Optimum for the locality

The Written Assignment – but also, a presentation!

Project stages (1)

Stage	When it happens	Who does it	Output
Identify a location, issue & relevant/reliable sources to support it	Wk 5, CS 8-9	You, with support from classmates and teacher	Draft version of project proposal: location, issue, working title
Check-in 1: explain & discuss your proposal; get feedback	Wk 6 tutorials	You and your teacher	2nd draft of proposal (if necessary) – to be incorporated into portfolio along with motivation/EDI
Submission of portfolio assessment: project proposal (in CV); motivation & EDI (in cover letter)	Wk 7 – **Monday 13th April**, 11 pm	You!	CV & cover letter (not only about the project!)

Project stages (2)

Stage	When it happens	Who does it	Output
Check-in 2: bring a source text to class for discussion. The source should relate to your chosen locality and issue.	Wk 7, CS 10	You, with classmates & teacher	Feedback re. suitability of the source; direction to continue researching location, issue & solutions for Written Assignment (WA)
Group interview assessment	Wk 8	You with another student (both interviewees)	Respond orally to questions about your project proposal, motivation & EDI (+ other Qs about your application)
Selecting further sources for WA (on location, issue and solutions); planning of WA	Wk 10	You, with support from classmates and teacher	Final selection of source texts; draft plan of WA; begin drafting WA

Project stages (3)

Stage	When it happens	Who does it	Output
Check-in 3: show & discuss your plan and selection of sources	Wk 11 tutorials	You with teacher	Feedback on plan; finalise plan and continue drafting WA
Submission of Written Assignment Assessment (including sources)	Wk 12 - **Tuesday May 19th, 11pm**	You!	Final version of WA. You must also upload your sources to Moodle.
Presentation Assessment	Wk 16	You with another student	Oral presentation on your project including reflection on partner's project

More about the Written Assignment

- Your work must be supported by appropriate references
- We provide **2 sources** on the topic of urban infrastructure
- Additionally, you must find and use:
  - 1-2 appropriate sources about your **chosen locality and its specific issue**
  - 1-2 appropriate sources about **each solution to the issue** (research **two** solutions)
  - ∴ Minimum of **5** sources, maximum of **8**
- Length = 1000 words + abstract of 100-150 words
- IEEE in-text referencing, reference list, & supply PDFs of sources

By the end of this week, you will have access to:

- The Written Assignment Tasksheet & Marking Criteria
- The two given WA sources (one lecture video, one book chapter)
  - ASU Engineering, *NAE Grand Challenges For Engineering - Restore and Improve Urban Infrastructure*, (Dec. 20, 2018). Accessed: Jan. 26, 2026. [Online Video]. Available: https://youtu.be/L-cPTYIOcFA?si=2GfC5tYJHYa2b5Cq
  - A. S. Permana, *Urban Engineering*, 1st edn. New York: Routledge, 2025. doi: 10.1201/9781032664613. [NB – chapter 1 only]

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Lecture 6: A closer look at the Grand Challenge

Restore and Improve Urban Infrastructure

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Aims

- Gain insights into the Grand Challenge “Restore and Improve Urban Infrastructure”
- Learn about urban infrastructure issues and solutions
- Develop listening skills in relation to a lecture on this topic

Task 1.1 Discuss and define

1. Which other NAE Grand Challenges do you think are connected with “urban infrastructure”?

2. How would you define “urban infrastructure”? Work together to come up with a one-sentence definition. (There is an image on the next slide to help you).

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Edward Kavazanjian, Jr., Ph.D., P.E., Dist.M.ASCE, NAE
Regents Professor and Ira A. Fulton Professor of Geotechnical Engineerir
School of Sustainable Engineering and the Built Environment
ASU Ira A. Fulton Schools of Engineering
Arizona State University

Task 1.1 Feedback

1. Dr Kavazanjian mentions:
   a. Provide access to clean water
   b. Develop carbon sequestration methods
   c. Make solar energy economical
   d. Manage the nitrogen cycle
   e. Provide energy from fusion
   f. Engineer the tools of scientific discovery
   g. Prevent nuclear terror

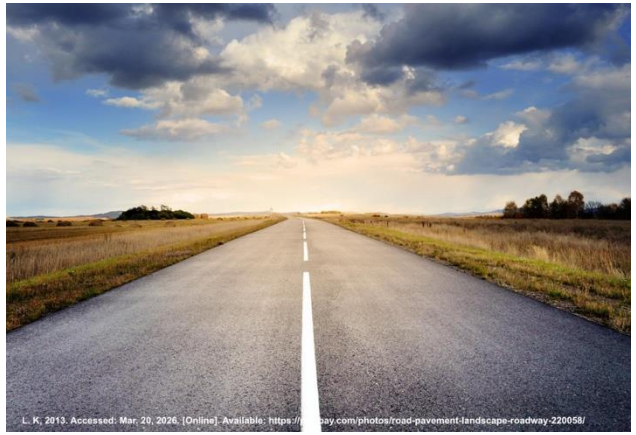
2. Urban infrastructure = "**The fundamental systems that a community relies upon to support the society that we live in** … transportation systems, water supply systems, power systems, housing, industry, business facilities, basically all of the systems that support our everyday life."

Task 2.1 Prediction and notetaking

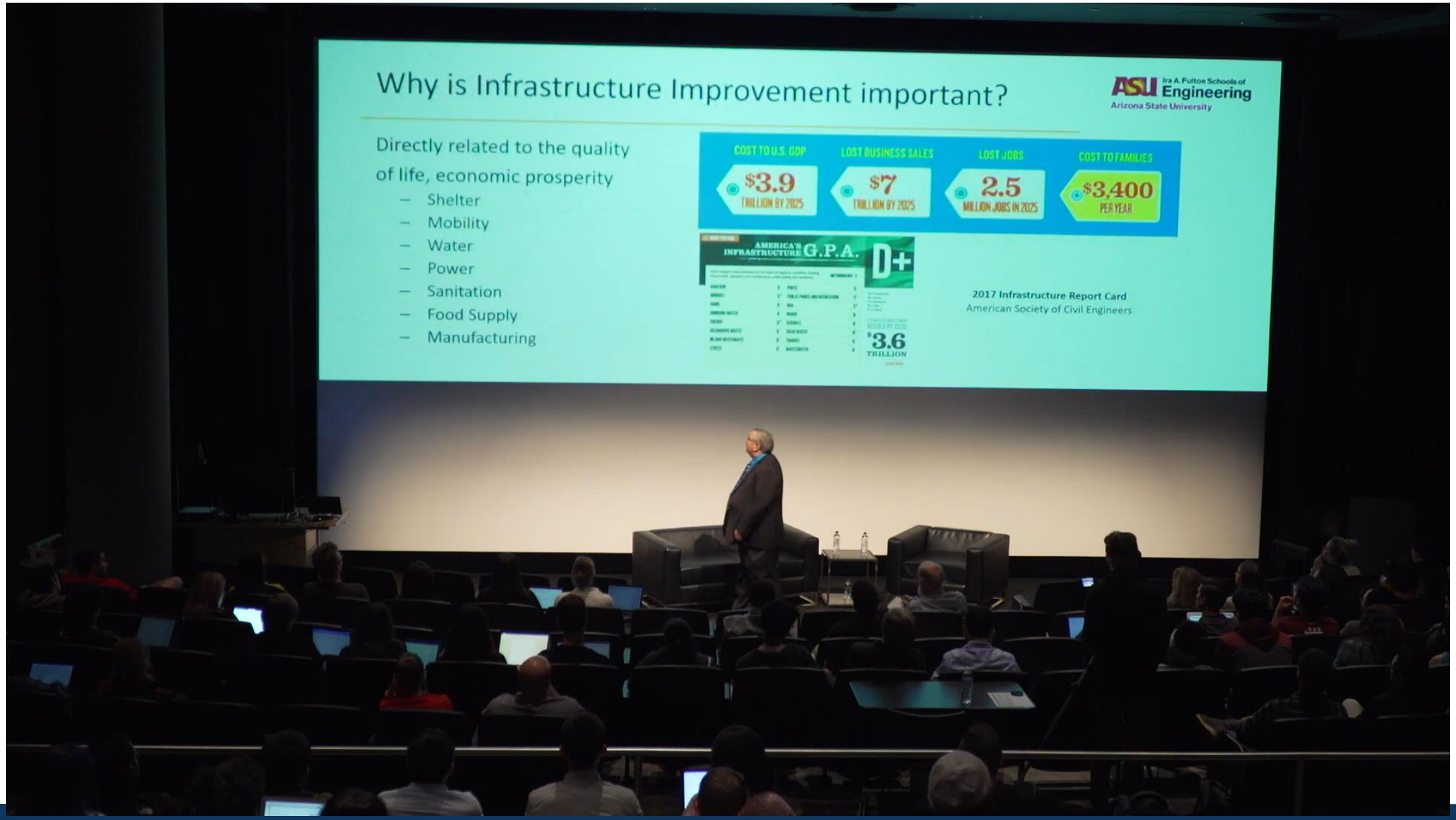
Look at the first of the three headings in your student materials.
"The importance of urban infrastructure and its condition in the USA"
Discuss with the student(s) next to you and make predictions in the first column. What do you think Dr Kavazanjian will mention?

Next, watch the first part of the main body of the lecture and take notes for this section in the second column.
After feedback, you will repeat the process for the 2nd and 3rd parts.

asphalt (n)



concrete (n)


Why is Infrastructure Improvement important?
ASU Ira A. Fulton Schools of Engineering
Arizona State University
Directly related to the quality of life, economic prosperity
– Shelter
– Mobility
– Water
– Power
– Sanitation
– Food Supply
– Manufacturing
COST TO U.S. GDP
$3.9
TRILLION BY 2025
LOST BUSINESS SALES
$7
TRILLION BY 2025
LOST JOBS
2.5
MILLION JOBS IN 2025
COST TO FAMILIES
$3,400
PER YEAR
AMERICA'S INFRASTRUCTURE G.P.A.
D+
$3.6
TRILLION
2017 Infrastructure Report Card
American Society of Civil Engineers

Task 2.1 selected feedback

The importance of urban infrastructure and its condition in the USA

- v. important for quality of life, prosperity
- condition in USA is poor – “D+” grade (= a fail in the US system)
- major investment needed; many systems in decay - maintenance deferred (delayed)
- ∴ many stresses on system, economic cost ($3,400 per year per family)
- also probs w/ water quality/poisoning – action not taken

History of civil engineering priorities in the USA

- pre-1950s – heroic structures
- 50s-70s – life cycle cost assessment (e.g. to choose mats)
- 70s-2000 – environ. impact assess. & mitigation
- 2000s – sust. & resilience (hazard mitigation) inc. social equity (≅ equality)
- now – climate ch. mitigation, sust. = “enough for everyone forever”, social equity

Task 2.1 selected feedback (cont.)

Urban infrastructure issues & solutions

- small % of earth, bt exhaust resources, generate pol.
- ∴ pay att. to env. & energy use; assess long-term impact ; tech dev.; new mats; mats with < carb. f/print
- mit. natural hazards, adapt. clim. change
- more eff. design & construct. processes
- extreme events – e.g. hurricanes, flooding, earthquakes
- “Anthropocene” – era of human-influenced climate & geosyst.
- more flooding in urban areas – paved with asphalt - <time of concentration

Out-of-class work

Homework:

- Review your notes and the PPT slides
- Read the Written Assignment tasksheet on Moodle
- Look at both source texts related to the WA on Moodle:
    - Review the lecture (with subtitles if needed) to check your answers to Task 2.1 in detail
    - Read the book chapter “Urban Engineering” before CS8.

Extra work:

- Watch the rest of the lecture and Q&A for additional ideas
- Use the Lecture Form on Moodle to ask any questions about the information covered in Lectures ˙ 5 & 6

NB: bring a laptop or tablet to class in Week 5!